with the final product to make software work more reliably, quickly and without hassles.

There is a wide range of open source machine learning frameworks available in the market, which enable machine learning engineers to build, implement and maintain machine learning systems, generate new projects and create new impactful machine learning systems.

Let's take a look at some of the top open source machine learning frameworks available.

Apache Singa

The Singa Project was initiated by the DB System Group at the National University of Singapore in 2014, with a primary focus on distributed deep learning by partitioning the model and data onto nodes in a cluster and parallelising the training. Apache Singa provides a simple programming model and works across a cluster of machines. It is primarily used in natural language processing (NLP) and image recognition. A Singa prototype accepted by Apache Incubator in March 2015 provides a flexible architecture of scalable distributed training and is extendable to run over a wide range of hardware.

Apache Singa was designed with an intuitive programming model based on layer abstraction. A wide variety of popular deep learning models are supported, such as feed-forward models like convolutional neural networks (CNN), energy models like Restricted Boltzmann Machine (RBM), and recurrent neural networks (RNN). Based on a flexible architecture, Singa runs various synchronous, asynchronous and hybrid training frameworks.

Singa's software stack has three main components: Core, IO and Model. The Core component is concerned with memory management and tensor operations. IO contains classes for reading and writing data to the disk and the network. Model includes data structures and algorithms for machine learning models.

Its main features are:

- Includes tensor abstraction for strong support for more advanced machine learning models
- Supports device abstraction for running on varied hardware devices
- Makes use of *cmake* for compilation rather than *GNU autotool*
- Improvised Python binding and contains more deep learning models like VGG and ResNet
- Includes enhanced IO classes for reading, writing, encoding and decoding files and data

The latest version is 1.0.

Website: *http://singa.apache.org/en/index.html*

Shogun

Shogun was initiated by Soeren Sonnenburg and Gunnar Raetsch in 1999 and is currently under rapid development by a large team of programmers. This free and open source toolbox written in C++ provides algorithms and data structures for machine learning problems. Shogun Toolbox provides the use of a toolbox via a unified interface from C++, Python, Octave, R, Java, Lua and C++; and can run on Windows, Linux and even MacOS. Shogun is designed for unified large-scale learning for a broad range of feature types and learning settings, like classification, regression, dimensionality reduction, clustering, etc. It contains a number of exclusive state-of-art algorithms, such as a wealth of efficient SVM implementations, multiple kernel learning, kernel hypothesis testing, Krylov methods, etc.

Shogun supports bindings to other machine learning libraries like LibSVM, LibLinear, SVMLight, LibOCAS, libqp, VowpalWabbit, Tapkee, SLEP, GPML and many more.

Its features include one-time classification, multi-class classification, regression, structured output learning, pre-processing, built-in model selection strategies, visualisation and test frameworks; and semi-supervised, multi-task and large scale learning.

The latest version is 4.1.0.

Website: *http://www.shogun-toolbox.org/*

Apache Mahout

Apache Mahout, being a free and open source project of the Apache Software Foundation, has a goal to develop free distributed or scalable machine learning algorithms for diverse areas like collaborative filtering, clustering and classification. Mahout provides Java libraries and Java collections for various kinds of mathematical operations.

Apache Mahout is implemented on top of Apache Hadoop using the MapReduce paradigm. Once Big Data is stored on the Hadoop Distributed File System (HDFS), Mahout provides the data science tools to automatically find meaningful patterns in these Big Data sets, turning this into 'big information' quickly and easily.

- *Building a recommendation engine:* Mahout provides tools for building a recommendation engine via the Taste library-- a fast and flexible engine for CF.
- *Clustering with Mahout:* Several clustering algorithms are supported by Mahout, like Canopy, k-Means, Mean-Shift, Dirichlet, etc.
- *Categorising content with Mahout:* Mahout uses the simple Map-Reduce-enabled naïve Bayes classifier.

The latest version is 0.12.2.

Website: *https://mahout.apache.org/*

Apache Spark MLlib

Apache Spark MLlib is a machine learning library, the primary objective of which is to make practical machine learning scalable and easy. It comprises common learning algorithms and utilities, including classification, regression, clustering, collaborative filtering, dimensionality reduction as well as lower-level optimisation primitives and higher-level pipeline APIs.